

CSE350: Digital Electronics & Pulse Techniques

Lecture 06: DTL Noise Margin, Fanout,
Modified DTL

Ex-1: If all inputs are high, then what is the magnitude of Noise Voltage that would cause the DTL NAND gate to malfunction?



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If we consider V_N amount of noise at any one input, say 'A', that might turn it on, then the voltage across D_1 must be at least 0.6V. So, we have,

$$V_{D_1} = V_P - V_A = 0.6. \text{ So, } V_A = 2.2 - 0.6 = 1.6V$$

$$\text{Now, } V_A = V_i + V_N$$

$$\text{Thus, } V_N = V_A - V_i = 1.6 - 5 = -3.4V$$

So, the magnitude of the noise voltage that can cause the DTL NAND to malfunction when both inputs are high is 3.4V.

This is called the **high threshold noise margin**, NM_H . So, $NM_H = 3.4V$

Note, that noise voltage can be either positive or negative, but the high threshold noise margin only considers the magnitude of noise voltage.

Ex-2: *If one input is high and the other is low, what magnitude of Noise Voltage at the low input terminal will cause the DTL NAND gate to malfunction?*

Like the previous example, we look back into Lec-05 & see that this is either of case(0,1) or case(1,0). Results will be same for both, so let's consider case(0,1). The diode D_1 at the low input terminal is on & D_2 is off at the high input terminal. So, it will **malfunction when D_1 turns off, misleading the circuit to operate as in case(1,1)**. The noise voltage V_N required to cause this can be found in the following manner:

D_1 has 0.7V across it when it's on. In case(0,1), D_3 , D_4 & T are off. Thus, $V_P - V_A = 0.7$

Or, $V_P = V_A + 0.7$. For low input with noise, $V_A = 0.2 + V_N$

So, $V_P = 0.9 + V_N$

Now, in the previous example, we saw that when D_3 , D_4 & T are on, $V_P = 1.7V$. So, the marginal value of, V_P that turns the transistor on into saturation, is 1.7V.

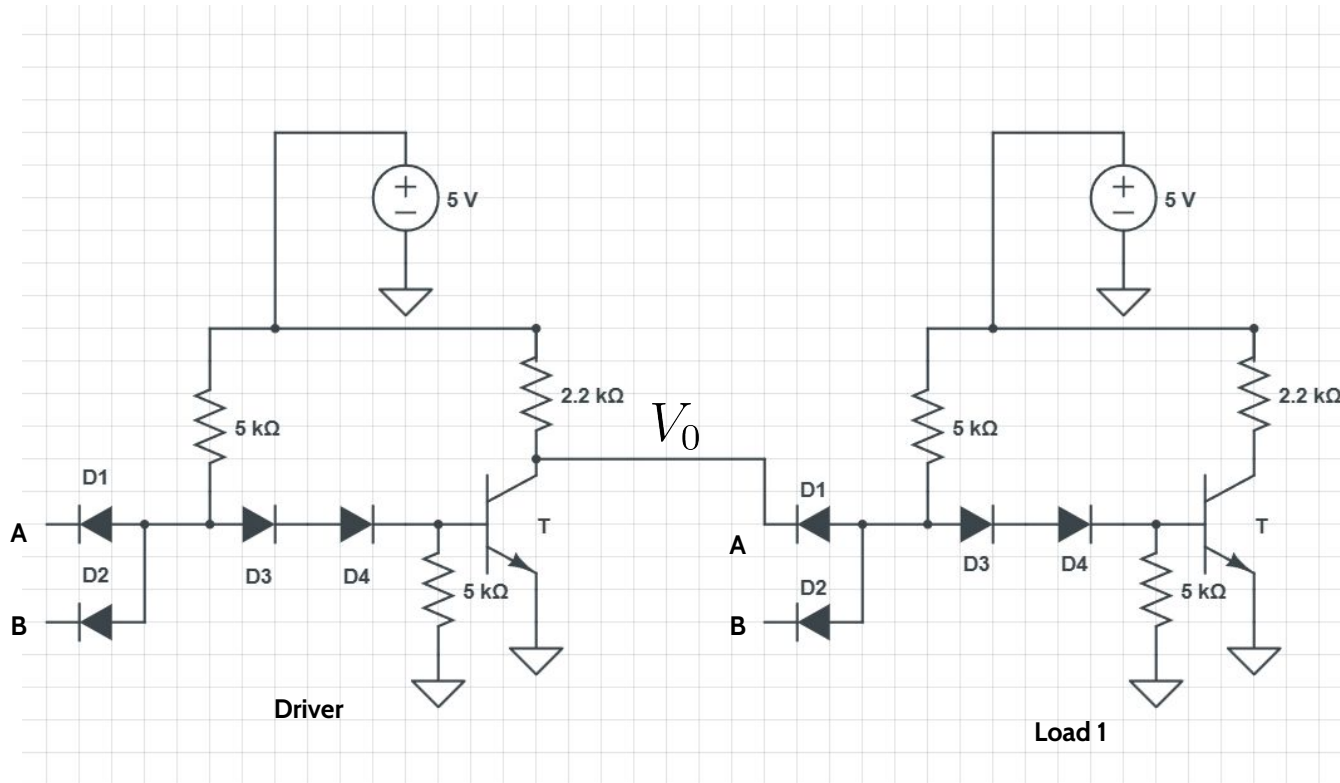
So, considering this marginal case, we get, $1.7 = 0.9 + V_N$

Or, $V_N = 0.8V$

The **low threshold noise margin**, $NM_L = 0.8V$

Maximum Fanout Calculation: It is also different from RTL circuit. Let's look at an example.

Ex-3: *If both the driver & load circuits are DTL NAND gates, calculate the maximum fanout.*



We start by considering the case when the **output voltage is high (case (0,1)/(1,0)/(0,0) of driver circuit)**. In this case, the transistor of driver, **T is in cutoff**. This leads to a **high voltage at the cathode of D_1** . Thus, the diode will be off & **no current will flow from driver to load**.

Consequently, the **maximum fanout in this case can be anything (i.e. infinity)**.

For the case when **output voltage is low**, the **diode at 'A' input will be on**. Thus, there will be current flowing from the 5V source of loads towards the driver circuit. Between the cases when 'A' is 0 [case (0,0), (0,1)], **maximum current flows through D_1 in case (0,1) when D_2 is off due to high voltage at 'B' of the load circuits**.

This current, say I_L can be found easily as in previous examples,
$$I_L = \frac{5 - V_P}{5k} = \frac{5 - 0.9}{5k} = 0.82mA$$

[Refer to Lec-05 on DTL NAND gate for details]

The amount of collector current that flows without these load currents,
$$I'_C = \frac{5 - 0.2}{2.2k} = 2.182mA$$

Say, maximum fanout is N. Then total collector current flowing through the transistor of driver circuit can be expressed as,
$$I_C = I'_C + N * I_L$$

Constraint here is that, the collector current cannot reach a value that will drive the transistor into forward active mode. The marginal condition between saturation and forward active as we saw in examples with RTL circuits was, $\beta_{forced} = \beta_F$. Let forward beta be 30. Now, in the example in Lec-05, we found, when T is on, base current

$$I_B = I_1 - I_2 = 0.4mA$$

$$\text{Now, } \beta_{forced} = \frac{I_C}{I_B}$$

So, the final marginal equation we get for calculating the maximum fanout of this case,

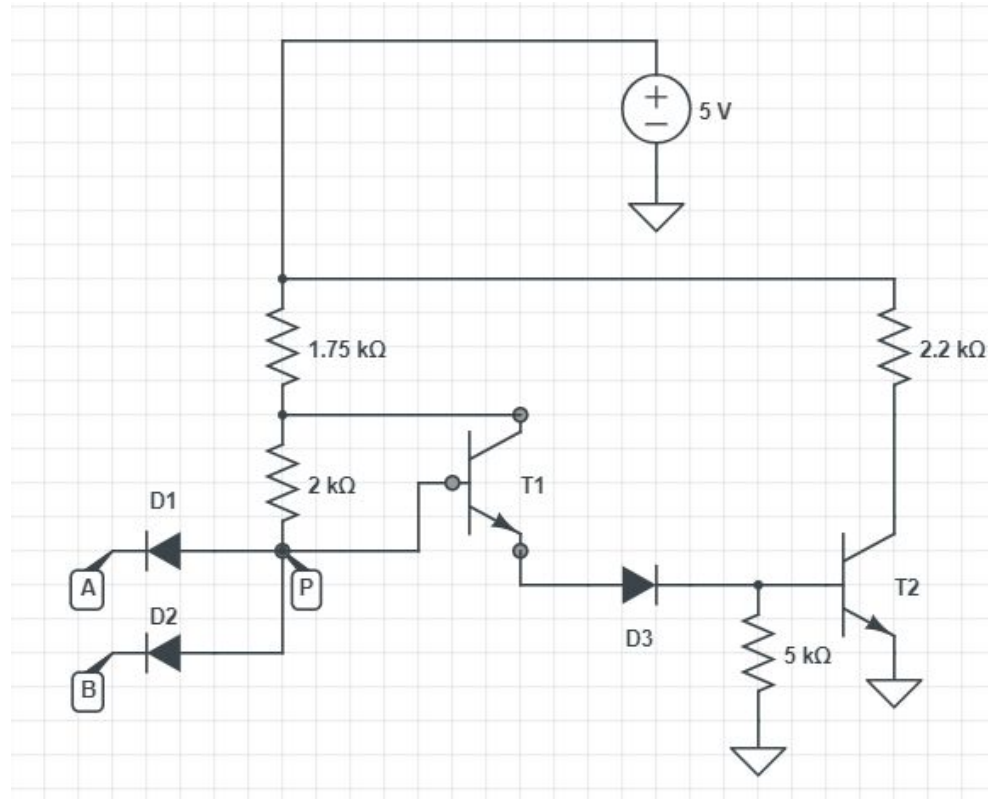
$$\frac{I'_C + N * I_L}{0.4} = 30$$

$$\text{or, } 2.182 + N * 0.82 = 30 * 0.4 = 12$$

$$\text{So, } N = \left\lfloor \frac{12 - 2.182}{0.82} \right\rfloor = \lfloor 11.97 \rfloor = 11$$

Thus, the overall fanout = $\min(11, \text{infinity}) = 11$

Modified DTL: The driver DTL NAND gate can be modified like the following, **to increase fanout**.



As we can see, instead of a diode, **an additional transistor is used at node 'P'**. Also, the resistance values are reduced. Due to this, **the current available from 5V will increase**, resulting in an enhanced fanout. However, the voltage at 'P' will be unchanged.

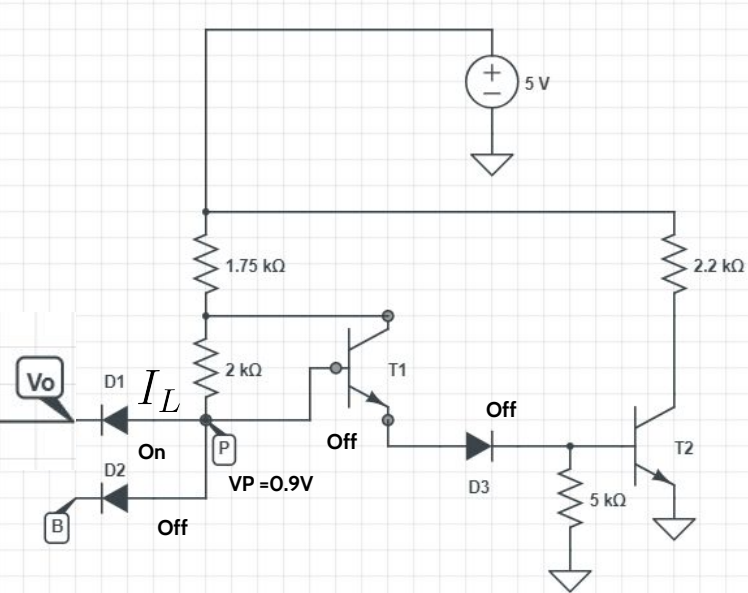
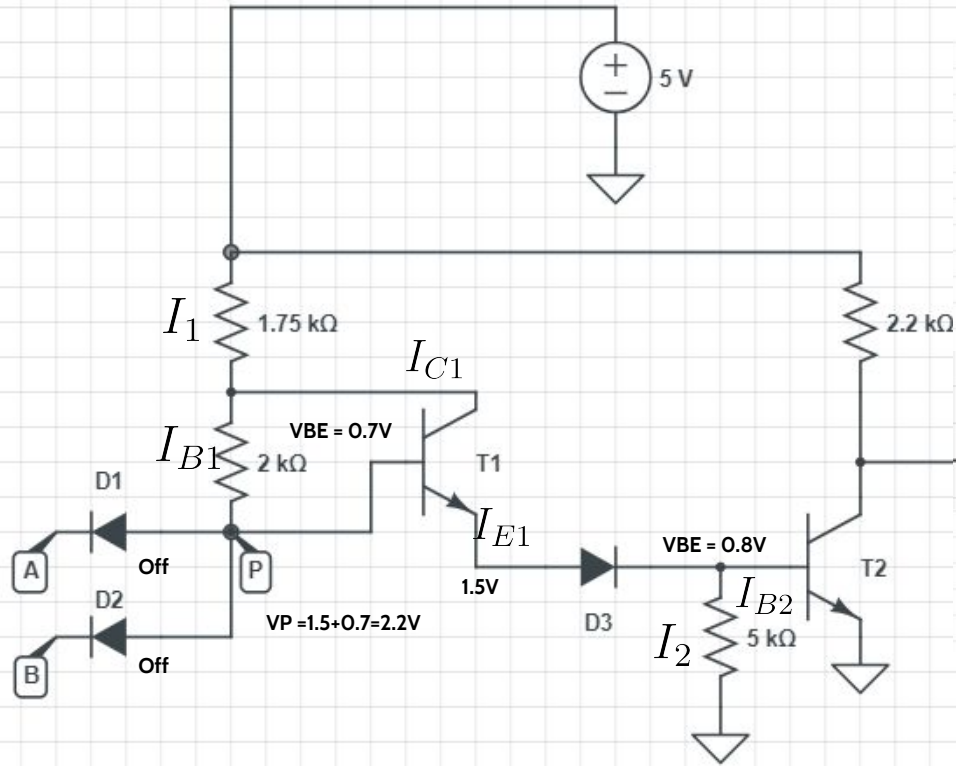
Let us analyze how this circuit increases fanout. The case of **high output voltage, will remain same since the driver transistor(T2) remains in cutoff**. So, maximum fanout is again infinity.

However, for the case of low output, driver transistor(T2) is in saturation mode. The input diode 'A' of load circuits is thus on & for maximum load current, we can again consider high voltage at 'B' of load circuit. Thus diode at 'B' will be off for load DTLs. This gives us the circuit in the following slide to analyze.

Here, as T2 will be in saturation, $V_{BE} = 0.8V$. The drop across diode, $D_3 = 0.7V$. Thus the emitter voltage of T1 = $0.8 + 0.7 = 1.5V$.

Since current flows from 5V to collector of T1 & then to base, $V_C > V_B \Rightarrow V_C - V_B > 0 \Rightarrow V_{CB} > 0$

So, $V_{BC} < 0$. **Also, since current is flowing through T1's terminals, it cannot be in cutoff. Thus only operating mode that can result in this condition is forward active.** So, T1 is in forward active mode. We know in F.A., $V_{BE} = 0.7V$. So, $V_P = 1.5 + 0.7 = 2.2V$



Applying KCL at the collector of T1,

$$I_1 = I_{B1} + I_{C1} = I_{E1} = (\beta + 1) * I_{B1}$$

$$I_{B1} = \frac{V_{C1} - V_P}{2k} = \frac{V_{C1} - 2.2}{2k}$$

$$I_{C1} = \frac{5 - V_{C1}}{1.75k}$$

Solving the above equations, $V_{C1} = 2.3V$

$$\text{Thus, } I_{E1} = (\beta + 1) * I_{B1} = 31 * \frac{2.3 - 2.2}{2k} = 1.55mA$$

$$I_2 = \frac{0.8}{5k} = 0.16mA$$

$$I_{B2} = I_{E1} - I_2 = 1.55 - 0.16 = 1.39mA$$

$$I_{C2_{max}} = \beta * I_{B2} = 30 * 1.39 = 41.7 = I'_C + N * I_L$$

$$\text{Solving, } N = \left[\frac{41.7 - 2.182}{0.82} \right] = 35$$

Load current calculation:

Since, the load DTL NAND gates will also be changed, load current will change too. But the **state of the transistors in load circuit will be same as before, in cutoff**. Thus $V_P = 0.9V$

$$\text{Consequently, } I_L = \frac{5 - 0.9}{1.75 + 2} = 1.092mA$$